

KAUSAR YETUNDE MOSHOOD

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EDUCATIONAL QUALIFICATION

Currently looking for summer internship 2027 in SWE, AI/ML Research Scientist

- 2025-Present** **PhD in Computer Science, Oregon State University, USA**
Current Research: Building LEST, a framework that tests whether natural-language safety claims in AI model specifications are verifiable by replacing static benchmark evaluation with a grammar-based fuzzing/mutation technique which automatically generates new test prompts, probing refusal/compliance behavior across self-harm, violent content and fairness & bias categories. [Source Code](#)
Expected graduation: Sep 2030(PhD)
- 2018-2023** **B.Engr in Computer Engineering, Oduduwa University Ipetumodu(OUI)**
Grade: First Class Honors, **CGPA: 4.87/5.0** (3.89/4.00)
University Rank: (1/ 3,675) -Best Graduating Student 2023, University Prize
- 2012-2018** WASSCE, Homat Comprehensive College- **Principal Distinction Award**

SKILLS AND INTERESTS

Technical Skills:Python, Software Design & System Architecture, Machine Learning, Data Structures and Algorithms, Natural Language Processing, Statistical Analysis & Evaluation

Program Analysis & AI Concepts: Fault Localization, Large-Scale Data Mining & Analysis, Graph-Based Learning (GNNs), Software Testing (Delta Debugging, Grammar-Based Fuzzing, Mutation Testing), LLM Safety Evaluation & Red-Teaming

Tools: Jupyter Notebook, Git, GitHub, VS Code, Firebase, Hugging Face, PyTorch, spaCy

Personal Skills: Teamwork and Collaboration, Time management, Creativity, Adaptability, Excellent Communication Skills

PROJECTS

LEST [GitHub](#): A framework that tests whether natural-language safety claims in LLM model specifications are verifiable, using grammar-based prompt fuzzing to auto-generate test prompts probing refusal/compliance

Defects4REST [GitHub](#): The first REST API defect benchmark consists of 607 real-world defects from 50 projects, enabling reproducible fault localization, testing and program repair research with 110 manually replicated for single-command buggy/patched checkout.

YECS [Link to dataset](#): a 120-hour, 100K-pair Yoruba-English code-switching speech dataset from 140 bilingual speakers, featuring validated transcriptions, language identification tags and stratified benchmark splits for low-resource ASR.

SpeakAbroad [Demo Video](#): an AI-powered roleplay platform that helps international students build confidence in real-world English communication and cultural interactions through interactive conversational practice.

COETER [Google PlayStore](#): Streamlines OUI Engineering student learning with past questions, course resources, real-time updates, and interactive communication, serving 300+ users.

Ecowaste [Demo Video](#): Empowers community engagement in waste management and recycling through user-reported dump sites and recyclable listings, streamlining collection for recycling firms.

SmartED [GitHub](#) is an education app aiding students in finding study partners and converting study material images to text using OCR, generating quiz questions through AI, offering points for more attempts, and delivering scholarship updates via News API

KausarBot [GitHub](#) - This project was an AI project designed to interact with people going through depression. The AI Chatbot is fun to chat with and has lots of riddles, games, and puzzles.

RESEARCH EXPERIENCE

- 2025-Present** **Graduate Research Assistant, Oregon State University**
- Improving fault localization in REST API systems by developing **RESTFL**, a novel structured IR approach combining BM25 ranking, artifact-aware field weighting, and multi-artifact parsing with REST-specific query modeling, ([Slides](#) | [Poster](#))
 - Conducted large-scale analysis of 11,313 GitHub issues to develop novel fault localization techniques for REST API systems, and contributed to **Defects4REST**, the first benchmark of 607 real-world defects from 50 projects organized into 6 defect types and 13 subtypes enabling reproducible evaluation of fault localization, testing, and program repair across 12 real-world systems. **16 hours/week**
- 2024 -Present** **AI Researcher, LyngualLabs**
- Contributed to the [WAXALNet African ASR Benchmark](#), evaluating 57 ASR models across 19 African languages and showing that compact fine-tuned models achieve substantially lower WER than larger zero-shot models [Link to Hugging Face, Paper](#)
 - Coordinated and engineered the collection of 100 hours of high-quality code-switched audio and transcribe datasets across 10+ diverse domains [Link to YECS dataset](#)
 - Research, design, and develop AI models to tackle code-switching, low-resource languages, and language diversity through experiments. **5 hours/week**
- 2022-2023** **Undergraduate Research**
- Thesis Title: Development of a mobile application to enhance E-learning(A case study for RACOET)
- Supervisor: Dr. (Mrs.) B.B Brahim

Duties: Reviewed relevant journals, wrote the research proposal, designed and formulated user Experience, and launched on Google Playstore with over 300+ users. **25 hours/week for 8 months**

SELECTED PUBLICATIONS

Ludwig, O., Anklesaria, N., Jin, Z., Pophale, S., **Moshood, K.Y.**, DeVore, C. J., Gill, B., Villareal, C., Teranishi, K., & Motwani, M. (2026). Kaizen: Metamorphic Fuzzing and Differential Testing for LLM-Translated HPC Applications. arXiv:2607.04058. [Paper](#) · [Code](#)

Babatunde, O. B., Olufemi, V. T., Bolarinwa, E. T., **Moshood, K. Y.**, & Emezue, C. C.(2026) YECS: A 120-Hour Community-Curated Yoruba-English Code-Switching Corpus, *Rebuttal Phase*, Interspeech 2026

Moshood, K., Olufemi, V., Babatunde, O., Bolarinwa, E., & Oluwademilade, W. (2026). Evaluating Yoruba text-to-speech systems for accessible computer-based testing in visually impaired learners. *In Proceedings of the 7th Workshop on African Natural Language Processing (AfricaNLP 2026)* (pp. 228–234). Association for Computational Linguistics.

Bolarinwa, E., Babatunde, O., Olufemi, V., **Moshood, K.** & Williams, O.. (2026). Closing the Gap in Low-Resource ASR: Leveraging Multilingual Models for Code-Switched Yoruba-English Speech. *DLI 2025 Research Track, in Proceedings of Machine Learning Research* 302:1-9 <https://proceedings.mlr.press/v302/bolarinwa26a.html>.

Babatunde, O. B., Olufemi, V. T., Bolarinwa, E. T., **Moshood, K. Y.**, & Emezue, C. C. (2025). Beyond Monolingual Limits: Fine-Tuning Monolingual ASR for Yoruba-English Code-Switching. *Seventh Workshop on Computational Approaches to Linguistic Code-Switching (CALCS 2025)*. <https://aclanthology.org/2025.calcs-1.3.pdf>

PROFESSIONAL EXPERIENCE

- 2024-2025** **Artificial Intelligence Research & Community Manager, Robotics and Artificial Intelligence Nigeria**
- Led AI research initiatives, hackathons and fostered collaborative community engagement to drive innovation among robotics and AI enthusiasts across 50 Nigerian tertiary institutions.
- 2022-2023** **Mobile Engineer & Project Manager- Alpha World-UAE (Dubai) / IT Services (Contract)**
- Tracked task progress, maintained schedules, and managed dependencies using project management tools, proactively addressing potential issues to ensure timely completion.
- 2020** **Automation and Machine Learning Intern, Robotics and Artificial Intelligence Nigeria(RAIN)**
- Participated in projects that involved automation, including projects like a traffic light system, a device to measure height, an automated pumping machine

SELECTED PRESENTATIONS & PUBLIC SPEAKING

RESTLocator: Fault Localization for defects in REST API Systems - Pacific Northwest Programming Languages and Software Engineering (PNW PLSE) workshop 2026, University of Washington, USA

‘Can Text-to-Speech Systems enable Inclusive Computer-Based Testing? An Evaluation of Yoruba TTS for Visually Impaired Learners’ - *WIML workshop @ NeurIPS 2025, San Diego, USA*

Speaker at 50+ RAIN-INNetwork events across 22 tertiary Institution in Nigeria (Jan2024- June 2025)

‘Let’s build a Fast ChatApp with Flutter and Firebase’ - *Flutteristas Global Conference (Nov 2023)*

‘TechWellness: Navigating Health Challenges in the Digital Age ’- *FlutterBytes Global Conference (Nov 2023)*

SELECTED ACHIEVEMENTS/ AWARDS

- 2023** **Overall Best Female Graduating Student, Leadership Prizes**
An Award donated to female student with the highest CGPA in the university by Vice Chairperson, Chief (Mrs.) Iyabo Adedoyin
- 2023** **Best Graduating Student, University Prizes:**
An Award donated to student with the highest CGPA in the university by Chancellor, Prof. R.A. Adedoyin.
- 2023** **Best Overall Performance, College of Engineering and Technology**
College Prizes adjudged to student who achieved the overall performance for contributions to the life of the university and college
- 2023** **Best Overall Performance, Department of Computer Engineering:** Student with the best overall performance in each programme
- 2023** **First Class Award-** For outstanding and exceptional academic performance
- 2023** **Event Maestro Award & Impact Amplifier Award** -An award recognition among 231 clubs for achievements in the Google Developer Groups on Campus, Sub Saharan Africa
- 2023** **Ist Runner Up Hackathon-** An inter_university hackathon among 21 teams, issued by Nigeria Associate of Computer Science
- 2021** **Most Innovative** -In recognition of academic excellence and Innovation-Nigeria University of Engineer Students
- 2020** **Second Runner Up** - Hult Prize competition Oduduwa University.
- 2018** **Principal Distinction Award**-An award for making distinction from JSS1 to SS3 2017/2018 academic session

SELECTED LEADERSHIP ROLES

- 2026** AWS Student Builder Group Leader, Oregon State University
- 2022-2023** Community Lead, Google Developer Groups on Campus, Oduduwa University Ipetumodu(OUI), Nigeria
- 2022-2023** Senate President, Nigerian Universities Engineering Student Association, OUI
- 2023** Chief Judge, National Aeronautics and Space Administration Challenge Ile-Ife
- 2022** Chief Judge, Hult Prize,OUI
- 2021-2022** Senate Clerk, Nigerian Universities Engineering Student Association, OUI.
- 2020-2021** Department Representative, Nigerian Universities Engineering Student Association, OUI.